

Smart Glove: Translating Gestures Into Text

Ishita Bhoir

line 2: *dept. name of organization*
(*of Affiliation*) line 3:
name of organization (of
Affiliation)
Pune, India
ishitavbhoir@gmail.com

Tanuja Naphade

line 2: *dept. name of organization*
(*of Affiliation*) line 3:
name of organization (of
Affiliation)
Pune, India
tanujanaphade12@gmail.com

Vaibhavi Shinde

line 2: *dept. name of organization*
(*of Affiliation*) line 3: *name*
of organization (of
Affiliation)
Pune, India
vaibhavisinde13@gmail.com

Sameer Talekar

line 2: *dept. name of organization*
(*of Affiliation*) line 3:
name of organization (of
Affiliation)
Pune, India
sameertalekar2007@gmail.com

line 1: 5th Given Name Surname

line 2: *dept. name of organization*
(*of Affiliation*) line 3:
name of organization
(*of Affiliation*) line
4: City, Country
line 5: email address or ORCID

line 1: 6th Given Name Surname

line 2: *dept. name of organization*
(*of Affiliation*) line 3: *name*
of organization
(*of Affiliation*) line
4: City, Country
line 5: email address or ORCID

Abstract— This paper presents the design and implementation of a wearable Smart glove that can sense finger movements, translate them into letters, and then display the output on a 16×2 LCD display. The glove consists of flex sensors and Arduino Uno for the hardware, signal conditioning and binary mapping for the software part. It goes into detail about the glove's architecture and working principle. Our system showcases a practical and low-cost working solution for gesturebased communication. Its ability to display letters based on finger movement aids people with hearing or speech impairment.

Keywords— Smart glove, Flex sensors, Arduino, Letter recognition, Nonverbal communication, Dynamic self calibration.

I. INTRODUCTION

Being able to communicate clearly plays a huge role in social inclusion and independence. Yet, millions of people worldwide face barriers due to hearing or speech impairments. This often leads to social isolation and limited access to jobs or other such opportunities. One way to solve this is to use sign language. The main issue with using sign language is that most people are unfamiliar with it. So while a person may convey their thoughts using signs, others may not be able to interpret or understand them.

To remedy this, gesture recognition technologies have become popular in recent years. But many of the existing systems are highly expensive and overly complicated. This paper introduces a simple and cost-effective Smart Glove that uses Arduino and flex sensors to map letters to their corresponding flex values. The proposed system uses binary logic to represent the state of each finger. A finger can either be straight (flex state = 0) or bent (flex state = 1). These states are used to map letters to corresponding flex states. Techniques such as dynamic self-calibration and exponential moving average filtering are used to measure accurate and stable sensor readings. These readings are then processed using Arduino. Then, the corresponding letter is displayed on the 16x2 LCD screen. The system also allows the stringing of recognized letters to form words, allowing for fluid communication between users.

Our approach provides a affordable and easy solution to close the communication gap that is present. The combination of binary logic and dynamic calibration along with signal

conditioning offers a low-cost yet working solution to the problem.

II. RELATED WORKS

Gesture recognition with the help of wearable devices has been studied a lot due to its application in helping speech-impaired people. Many papers have studied the use of flex sensors, IMU sensors, and machine learning algorithms to guess hand gestures.

Filipowska et al. [1] proposed a glove that uses machine learning technology for gesture detection and recognition. This showed that sensor-based gloves could detect complex hand movements and translate them into text.

Telluri et al. [2] also presented a low-cost gesture recognition glove. The paper showed that flex sensors along with machine learning or simple binary mapping, can detect dynamic hand movements.

To improve the stability of flex values, Choi et al. [3] focused on signal conditioning techniques, explored applying median filtering and moving average smoothing to reduce noise from sensor readings. This significantly improved the accuracy of detecting bent and straight finger positions.

Sümbül et al. [4] developed a glove that combined MEMS and flex sensors with deep learning models. In contrast, Lakshmi et al. [5] implemented an Arduino and flex sensor-based glove for hand gesture recognition to assist the speech-impaired. Then El Rwelli et al. [6] focused on sign language recognition using convolutional neural networks. Although their approach managed to achieve robust gesture recognition, it required a lot of computational power and training data. This limited its real-time deployment on simple microcontroller platforms.

Many others have also worked on a smart glove [7]-[15]

On the other hand, the Smart Glove combines dynamic calibration, median filtering, EMA smoothing, and binary mapping for a real-time, noise-resistant letter recognition system with minimal hardware requirements. This approach combines the simplicity of binary mapping with robust signal conditioning, making it a practical and cost-effective solution for gesture-to-text translation in lowcost, portable systems.

III. PROPOSED SYSTEM

The proposed system is designed to convert finger movement into the corresponding alphabet using an Arduino Uno based glove prototype. The system uses five flex sensors (one along each finger) to capture finger movement, processes the analog signals through a filter, and displays the recognized letters in real time on a 16×2 LCD module. Our model is divided into hardware and software components as described below.

A. Hardware Components

The hardware of the glove consists of an Arduino Uno, flex sensors, a 16×2 LCD screen, 10kΩ resistors, and jumper wires. All the components are then fixed on a glove.

Each flex sensor acts as a variable resistor. The resistance changes as the finger bends or straightens, and this results in a variation in the analog voltage. The analog signals are connected to the Arduino's analog input pins (A0–A4).

The Arduino acts as the central processing unit. It processes the signals received from the flex sensors. Then the signal from flex sensors is converted from analog to digital. The Arduino Uno then filters the signal, performs letter mapping and LCD interfacing. The 16×2 LCD provides realtime output by showing the detected letter on the first row and the accumulated word or sentence on the second. Power is supplied through a 5V DC input usually using a USB cable connected to a device like laptop. The complete hardware specifications are listed below in Table I.

Table I: Hardware Specifications of the Smart Glove

Component	Specification
Microcontroller	Arduino Uno
Flex Sensors	5 units, 10kΩ base resistance
Display	16 × 2 LCD
Filtering Method	Median + EMA filtering
Power Supply	5 V DC (using USB)
Programming Tool	Arduino IDE
Communication	Serial 9600 bps (for testing)

The glove system was initially designed on Tinkercad. The simulation displays the hardware connections between the Arduino Uno, flex sensors, LCD, and supporting components. This allowed us to check if the circuit was correct before the actual implementation. The simulated circuit layout is shown in Fig. 1.

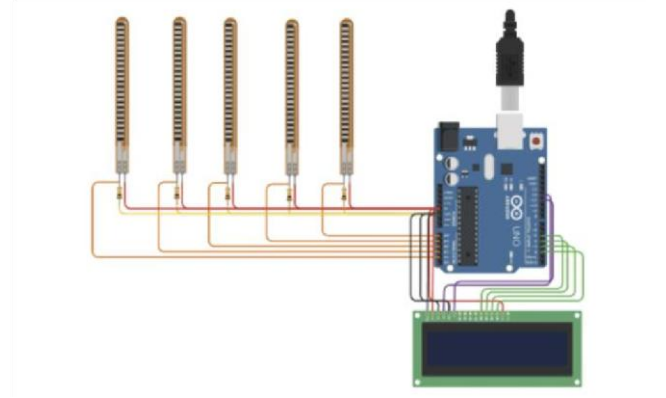


Fig. 1: Tinkercad Circuit Diagram

The block diagram of the smart glove is shown in Fig. 2.

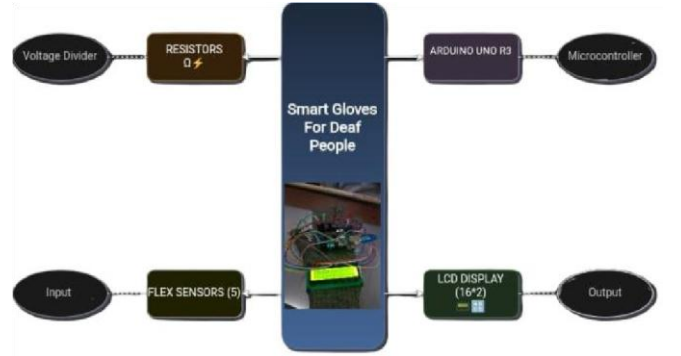


Fig. 2: Block diagram of the Smart Glove

B. Software Components

The software of the Smart glove is developed using Arduino IDE. It is based on two core logics: Signal Conditioning and Binary Mapping. Firstly, signal conditioning is used to filter out noise and enhance the accuracy of the readings. The second logic that we use is binary logic that helps map the flex values to the corresponding letter.

In the signal conditioning stage the data from flex sensors is passed through various techniques like median filtering, exponential moving average (EMA) smoothing, and dynamic calibration, etc. These techniques help stabilize the values. Signal conditioning techniques have shown to improve correctness in gesture recognition [3].

1) *Median Filtering*: This Median filter is used to remove random noise by taking three continuous samples. The median (M) is then selected using the given formula:

$$x_M = M(x_1, x_2, x_3) \quad (1)$$

This ensures random spikes or changes do not affect the bend state detection and stabilizes the value.

2) *Exponential Moving Average*: The filtered signal is then stabilized using the EMA technique. It smooths the rapid changes while still keeping its responsiveness. The EMA is calculated as:

$$EMA_{new} = \alpha \times V_{current} + (1 - \alpha) \times EMA_{previous} \quad (2)$$

where $\alpha = 0.4$ is the smoothing factor.

3) *Dynamic Calibration*: At the start, each flex sensor automatically sets its baseline (straight-finger state). A dynamic threshold is then computed according to:

$$\text{Margin} = \text{Fixed Margin} + (\text{Noise} \times K) \quad (3)$$

where K is the noise compensation coefficient. The dynamic calibration makes sure that the values are stable.

Using these methods together helped us reduce noise and helped give more stable readings for gesture detection.

Each flex sensor is assigned a variable representing a single bit for binary mapping. The sensors are labeled F1 through F5 starting from the thumb as F1 to the little finger as F5. The thumb i.e. F1 serves as the least significant bit (LSB) and F5 as the most significant bit (MSB). Each flex sensor can have one of two states: bent (1) or straight (0).

Each different combination of these five states forms a 5-bit binary code which is then mapped to a specific letter. For example, when only the thumb (F1) is bent and all other fingers (F2–F5) are straight, the binary code is 00001, which corresponds to the letter A. Table II provides additional examples of these binary codes and their corresponding letters.

Table II: Binary Mapping of Flex Sensors to Alphabet

Binary Code	Letter	Binary Code	Letter
00001	A	01110	N
00010	B	01111	O
00011	C	10000	P
00100	D	10001	Q
00101	E	10010	R
00110	F	10011	S
00111	G	10100	T
01000	H	10101	U
01001	I	10110	V
01010	J	10111	W
01011	K	11000	X
01100	L	11001	Y
01101	M	11010	Z

C. Working Principle

We noticed the glove's behaviour followed three steps. First, it sensed any finger bend, then the Arduino processed it, and finally, the LCD showed the letter.

1) *Sensing*: The flex sensors sense any movement in fingers, like bending.

2) *Processing*: The Arduino board takes the analog signal and converts it to a digital signal. Then the Arduino applies signal conditioning to filter out and stabilize the signal. It is then mapped to the corresponding letter and sent to the LCD to display.

3) *Output*: The resulting letter is displayed on the 16×2 LCD.

The glove translates finger movements into readable text. This demonstrates a low-cost and working solution for gesture-based communication.

D. Prototype

In the initial stage, the logic of signal conditioning and binary mapping was tested using a single flex sensor along with Arduino and a 16×2 LCD screen using a breadboard as shown in Fig. 3. Using a single flex sensor, Letter A or B was displayed on the screen, depending on whether the flex sensor was bent or straight, as shown in Fig. 4.

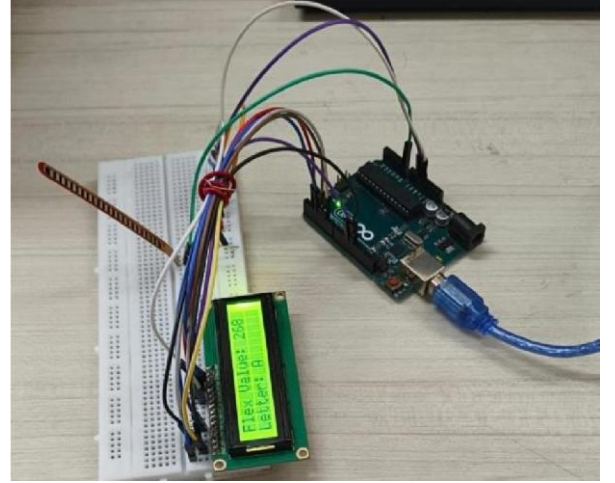


Fig. 3: Prototype with a single flex sensor on a breadboard (photo taken by author).



Fig. 4: Prototype with a single flex sensor displaying the accurate letter (photo taken by author).

After it was tested with one flex sensor, the prototype was tested by incrementally expanding the number of flex sensors. Once all flex sensors were added and tested on the breadboard, they were assembled on a glove to capture complete hand gestures. The five flex sensors allowed the full binary mapping logic to be implemented. It was able to convert combinations of bent and straight fingers into letters in real time. The final glove prototype is shown in Fig. 5.

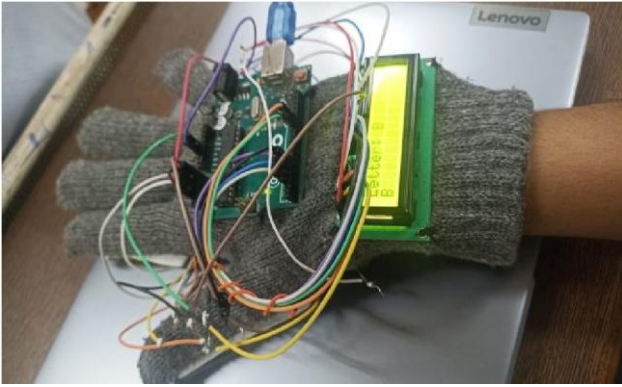


Fig. 5: Final glove prototype (photo taken by author).

This staged prototyping process ensured system reliability and accuracy, while also providing troubleshooting at each stage of development. The Glove could then display the corresponding letter when a finger was bent, as shown in Figs. 6-7.

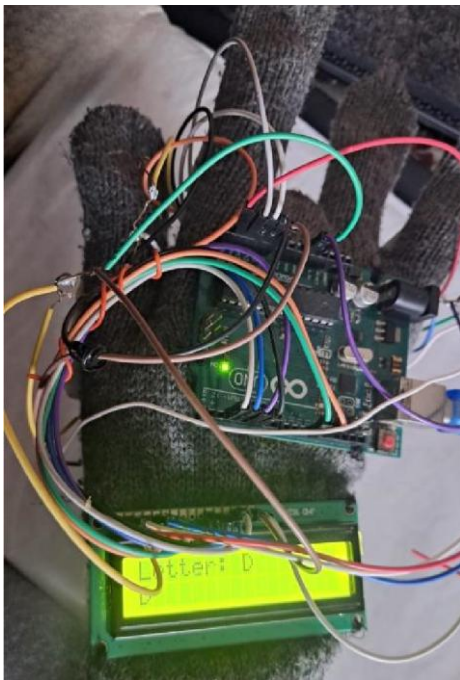


Fig. 6: Letter D being displayed when middle finger is bent (00100).



Fig. 7: Finger being bent

The Smart Glove can string together letters to form words, as shown in Figs. 8–11.



Fig. 8: Output displayed for the word "OK"

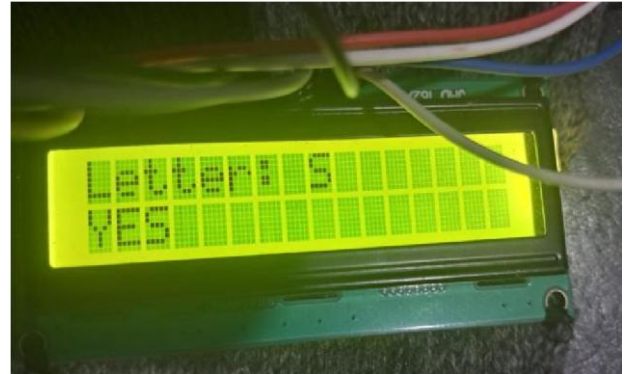


Fig. 9: Output displayed for the word "YES"



Fig. 10: Output displayed for the word "NO"



Fig. 11: Output displayed for the word "HELP"

IV. RESULTS AND DISCUSSION

The Smart Glove was tested in two stages: first using a single flex sensor and then with the complete fivefinger prototype. Using a single sensor, the effectiveness of the signal conditioning techniques was checked. When it was working accurately, we increased the number of flex sensors. We used dynamic calibration to adapt to variations

in flex values each time the glove was used. It also accounted for different sensor placement and finger length.

For the full five-finger prototype, each sensor (F1–F5) was mapped to a 5-bit code for letter recognition. The 16×2 LCD displayed the current detected letter on the first row and the formed word on the second row. Most letters were recognized correctly, but a few needed slower finger movements to stay accurate. During testing, some unexpected readings appeared due to loose sensor contact. We fixed this by rechecking every connection again.

During testing we tried each letter twenty times to find the accuracy. The result of that testing is given below.

Table III: Accuracy Analysis

Letter	Correct Detection (Out of 20)	Accuracy (in %)	Letter	Correct Detection (Out of 20)	Accuracy (in %)
A	18	90	N	16	80
B	20	100	O	11	55
C	18	90	P	20	100
D	20	100	Q	14	70
E	17	85	R	18	90
F	19	95	S	15	75
G	16	80	T	16	80
H	20	100	U	13	65
I	17	85	V	15	75
J	18	90	W	12	60
K	16	80	X	17	85
L	17	85	Y	13	65
M	14	70	Z	13	65

As observed from the above table, the letters using single fingers were most accurately detected. The increase in the number of fingers used caused the accuracy to decrease a little. The letters which used the sensor attached to the thumb were found to be a little less accurate. The average accuracy of recognition of all the letters is found to be around 81.35% (Low performing letters like O bring the average down significantly).

Table IV: Comparison with previous technology

System/ Paper	Technology Used	Accuracy	Hardware Complexity
Filipowska et al. [1]	ML Based Smart Glove	90	High
Proposed System	Arduino + Flex Sensor	81.35	Low

The above hardware complexity comparison is qualitative and based on computational requirements and sensor integration.

Overall, the system demonstrates reliable and low-cost gesture-to-text translation. The signal conditioning methods effectively reduced noise and improved accuracy, while the binary mapping logic enabled consistent letter recognition. We observed certain limitations, like reduced accuracy for incomplete finger bends. Our project also showed an inability to support dynamic gestures or use by multiple users without recalibration. These results provide areas for future improvement.

V. CONCLUSION

The finger movements are translated into letters using the flex sensors. Thus, our glove is simple and affordable while still being in working state.

In the initial stages of testing with a single flex sensor, we noticed the effectiveness of the signal conditioning and the binary mapping logic. Then the full five-finger prototype made sure that the glove can display letters and string them together to form words. From the results, we can say that the system shows good accuracy.

While the current prototype performs correctly, some limitations still exist. The limitations include reduced accuracy for incomplete finger bends and they could not handle dynamic gestures. These limitations provide scope for future improvement. It also provides scope for implementing different mapping algorithms, wireless data transmission or using machine learning models.

So, our Smart Glove provides an affordable and reliable solution for gesture-to-text translation that works by combining simple hardware with signal conditioning and binary mapping. It is reliable as well as useful.

REFERENCES

[1] A. Filipowska, W. Filipowski, P. Raif, M. Pieniążek, J. Bodak, P. Ferst, *et al.*, "Machine learning-based gesture recognition glove: Design and implementation," *Sensors*, vol. 24, no. 18, p. 6157, Sept. 2024. [Online]. Available: <https://doi.org/10.3390/s24186157>.

[2] T. Telluri et al., "Low-Cost Flex-Powered Gesture Detection System and Its Applications," ResearchGate, May 2025.

[3] S. Choi et al., "Noise Reduction in Glove-Based Gesture Recognition Systems Using Median and Moving Average Filters," *Sensors and Actuators A: Physical*, vol. 150, pp. 123–130, 2009.

[4] H. Sümbül et al., "A Novel MEMS and Flex Sensor-Based Hand Gesture Recognition and Regenerating System Using Deep Learning Model," *Journal of Electronics and Communication Engineering*, vol. 9, no. 2, pp. 1–10, 2025.

[5] S. Lakshmi et al., "Arduino and Flex Sensor-Based Hand Gesture Recognition for Speech Impaired People," *International Journal of Engineering Research & Technology (IJERT)*, vol. 3, no. 5, pp. 1–5, 2014.

[6] R. El Rwelli et al., "Gesture-Based Arabic Sign Language Recognition for Impaired People Based on Convolution Neural Network," *arXiv preprint arXiv:2203.05602*, 2022.

- [7] M. P. Verma, S. L. Shimi, and S. Chatterji, "Design of Smart Gloves," *Int. J. Eng. Res. Technol.*, vol. 3, no. 9, pp. 639–642, Sept. 2014.
- [8] Z. R. Saeed, Z. B. Zainol, B. B. Zaidan, and A. H. Alamoodi, "A systematic review on systems-based sensory gloves for sign language pattern recognition: An update from 2017 to 2022," *IEEE Access*, vol. 10, pp. 104236–104260, 2022. [Online]. Available: <https://doi.org/10.1109/ACCESS.2022.3219482>
- [9] N. Praveen, N. Karanth, and M. S. Megha, "Sign language interpreter using a smart glove," in *Proc. 2014 Int. Conf. Adv. Electron. Comput. Commun. (ICAECC)*, Bangalore, India, Oct. 10–11, 2014, pp. 1–5. [Online]. Available: <https://doi.org/10.1109/ICAECC.2014.7002401>
- [10] W.-C. Chuang, W. J. Hwang, T. M. Tai, D.-R. Huang, and Y.-J. Jhang, "Continuous Finger Gesture Recognition Based on Flex Sensors," *Sensors*, vol. 19, no. 18, p. 3986, 2019.
- [11] M. A. Ahmed, B. B. Zaidan, A. A. Zaidan, M. M. Salih, and M. M. bin Lakulu, "A Review on Systems-Based Sensory Gloves for Sign Language Recognition: State of the Art between 2007 and 2017," *Journal of Engineering Science and Technology Review*, vol. 10, no. 2, pp. 8–25, 2017.
- [12] A. J. Abougarair and W. A. Arebi, "Smart Glove for Sign Language Translation," *International Robotics & Automation Journal*, vol. 8, no. 3, pp. 109–117, 2022.
- [13] M. S. Amin, S. T. H. Rizvi, and M. M. Hossain, "A Comparative Review on Applications of Different Sensors for Sign Language Recognition," *IEEE Access*, vol. 10, pp. 34650–34664, 2022.
- [14] P. Sharma and R. Gupta, "Design and Implementation of Sensor-Based Smart Glove for Sign Language to Speech Conversion System," *Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 11, no. 4, pp. 707–712, Apr. 2024.
- [15] S. Panda and S. Chakravarty, "Hand Gesture Recognition Using Flex Sensor and Machine Learning," in *Proceedings of the International Conference on Computational Intelligence and Data Science (ICCIDS)*, pp. 1–6, 2023.